

ALEXANDER M. VANCE

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PROFESSIONAL SUMMARY

Accomplished and mission-driven Senior Data Scientist with over 6 years of progressive experience executing advanced statistical modeling, predictive analytics, and automated machine learning (ML) workflows within regulated financial environments. Proved expertise leading data science initiatives, developing production-level analytics pipelines, and delivering strategic technical guidance to executive leadership. Deep technical knowledge of market microstructures, algorithmic trading datasets, and emerging asset classes including crypto assets and decentralized event/prediction markets. Expert in translating complex computational discoveries into actionable regulatory strategy, market surveillance capabilities, and policy risk scoring frameworks.

CORE COMPETENCIES & EXPERTISE

- **Data Science Governance & Strategy:** End-to-end ML lifecycle management, technical pipeline architecture, regulatory compliance metrics, and technical staff mentorship.
- **Computational Analytics & Modeling:** Advanced statistical modeling (logistic, tree-based, ensemble methods), anomaly detection, automated exploratory data analysis (EDA), and natural language processing (NLP) for unstructured financial disclosures.
- **Big Data & Cloud Infrastructure:** Distributed computing architectures, cloud-based analytics platforms (AWS/Azure), SQL/NoSQL databases, and automated ETL/data mining workflows.
- **Financial Market Surveillance:** Time-series trading data engineering, risk-scoring framework construction, order-book anomaly detection, and market manipulation discovery.

PROFESSIONAL EXPERIENCE

Senior Data Scientist (Equivalent to GS/CT-12)

Capital Markets Regulatory Authority / Financial Analytics Division – Washington, DC

03/2023 – Present | 40 Hours/Week

Salary: \$138,500 | **Supervisory/Lead Status:** Technical Lead & Mentor

Strategic Data Science Advisory & Project Leadership

- **Lead Project Delivery:** Serve as the Principal Data Scientist leading a cross-functional squad of 5 data engineers and analysts to design and deploy the agency's next-generation automated surveillance engine for high-frequency and derivatives market activity.
- **Executive Briefings & Technical Guidance:** Provide direct technical counsel to the Chief Data Officer and senior enforcement attorneys on data-driven patterns of market anomalies. Translate complex predictive modeling outputs into clear executive briefs and interactive visual dashboards, enabling rapid-response policy updates and rule-generation.
- **Staff Mentorship:** Actively mentor junior data scientists and researchers, establishing standard operating procedures for code reviews, testing frameworks, and reproducible algorithmic pipelines.

Advanced Computational Analytics & Infrastructure Engineering

- **Automated Data Workflows:** Architected and implemented end-to-end automated data science pipelines using R and Python, condensing processing times for unstructured regulatory filings and structured transaction data by 45%.
- **Machine Learning & Risk Scoring:** Developed and operationalized supervised and unsupervised machine learning architectures (including robust tree-based ensembles and advanced generalized linear/logistic models) to flag suspicious trading behaviors and insider trading signs in emerging decentralized spaces.
- **Exploratory Data Analysis (EDA):** Spearheaded massive scale data mining initiatives across multi-terabyte data warehouses, consolidating complex statistical summaries—such as multi-variable distributions, boxplots, and anomalies—into centralized analytical layers to support enforcement investigations.
- **Cloud Architecture Migration:** Orchestrated the migration of legacy financial modeling tools onto a secure cloud-based analytics platform, enhancing processing capabilities for high-throughput market data while maintaining strict governance, data integrity, and compliance standards.

Data Scientist / Quantitative Analyst

Apex Quantitative Derivatives Group – Chicago, IL

08/2020 – 02/2023 | 40 Hours/Week

Salary: \$122,000 | **Supervisory/Lead Status:** Project Co-Lead

Data Infrastructure & Predictive Modeling

- **Predictive Model Development:** Built and optimized advanced time-series and predictive statistical models to analyze liquidity shifts and market depth across emerging market segments and traditional commodity futures.

- **Data Quality & Extraction:** Engineered robust data mining and cleaning protocols to ingest, validate, and parse high-frequency tick data from diverse, fragmented data structures. Ensured 99.9% data integrity across structured order books and unstructured text-based market sentiment data.
- **Risk Mitigation Engineering:** Created an internal algorithmic risk-scoring system using automated data pipelines, reducing false-positive anomaly alerts by 30% and significantly streamlining the firm's compliance workflows.
- **Cross-Functional Collaboration:** Partnered closely with data infrastructure engineers to design scalable database schemas capable of handling billions of rows of historical market records, facilitating rapid mathematical back-testing of trading patterns.

EDUCATION

Master of Science in Data Science & Statistical Computing

Northwestern University – Evanston, IL | Graduated: 2020

- **Focus Areas:** Advanced Statistical Modeling, Machine Learning Architectures, Big Data Infrastructure, Time-Series Analysis.

Bachelor of Science in Mathematics and Statistics

University of Illinois at Urbana-Champaign – Champaign, IL | Graduated: 2018

- **Honors:** Magna Cum Laude

TECHNICAL SKILLS TOOLKIT

- **Programming Languages:** R (caret, tidyverse, ggplot2, MASS), Python (pandas, scikit-learn, NumPy), SQL, Bash.
- **Big Data & Cloud Platforms:** AWS (S3, EC2, EMR, Athena), Apache Spark, Databricks, Docker, Git/Version Control.
- **Analytics & BI:** Tableau, Shiny, PowerBI, Automated Markdown Reporting.
- **Data Environments:** PostgreSQL, MongoDB, Parquet/Hadoop Distributed File Systems (HDFS).

FEDERAL KSA COMPLIANCE MAPPING

This resume explicitly demonstrates and satisfies the required core competencies for the **CT-13 Data Scientist-Emerging Markets** position:

- **Computational Analytics & Technology Evaluation:** Proved via 6+ years of building advanced statistical, logistic, and machine learning models to capture trends and anomalies across massive structured and unstructured datasets.
- **Data Infrastructure & Mining:** Validated by extensive history engineering scalable cloud-based pipelines (AWS), optimization of data extraction, and managing rigorous data cleaning protocols for multi-terabyte financial databases.
- **Data Science Governance & Strategy:** Illustrated through professional experience establishing reproducible analytical environments, standard operating frameworks, and maintaining alignment with broad enterprise data strategies.
- **Strategic Data Science Advisory & Systems Development:** Documented via a proven record of leading technical data teams, mentoring staff, and translating highly complex computational discoveries directly into executive recommendations for CDOs and senior agency directors.